Lathe Final Report

2.720 Elements of Mechanical Design – Spring 2025

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The following table summarizes the measured performance of the constructed lathe in meeting functional requirements set by the team to meet course objectives and satisfy our model customer of a hobby roboticist. The appendix shows experimental procedures and detailed results for these requirements for which measurement was non-trivial.

#	Requirement Name	Allowable Range	Measured Value
1	Repeatability	50 micron +0, -40	31 micron
2	CNC Z Travel Accuracy	2 micron/mm +3, -1.5	(1-6.1) micron error / mm
3	Swing Radius	37.5 mm +/-12.5	40 mm
4	Maximum Part Length	160 mm +140, -60	101 mm
6	Power Draw	900 W (max)	330 W
7	CNC X and Z Resolution	15 micron +/-5	20 micron
8	Weight	40 lbs +/-20	41 lbs
9	Actuation torque required	2 Nm +2, -1.5	< 0.3 Nm
10	Envelope (depth x width x height)	600 x 2200 x 400 mm	343 x 495 x 353 mm
11	Cost of materials not provided	$250 (max)	$231.22

Summary of Testing and Optimization

The perpendicularity of the X and Z drive axes was determined using a Coordinate Measuring Machine (CMM) to find the location of datum spheres mounted in place of the tool-post. This was used to inform the adjustment of the dovetail-carriage bolted connection. After tuning, the X and Y drives were found to be within 0.27 degrees from perpendicular. See Appendix for details.

While not strictly a functional requirement, the theoretical material removal rate (MRR) for 4130 steel was 0.3 in^3/min and heavy cuts in this materials was measured to be 0.16 in^3/min by turning down a 0.61" diameter steel cylinder using 0.080" depth of cut and 2.15 in/min feed rate. This resulted in a 1100 rpm spindle speed and 19.6 amp current draw. We also achieved a 0.19 in^3/min MRR by turning down a 5/8 " diameter steel stock by putting a carbide tool at the center of the piece and removing material as fast as possible.

Repeatability was measured by locking the X axis in a position to make a 50 thou radial depth cut for a ½” hex steel stock. Five pieces were installed into the lathe and turned down approximately ¼” in length to take micrometer (clutch) measurements. Given the X axis was locked, we removed potential human-error in not making the same depth of cut between artefacts.

Runout was measured using the Donaldson Reversal method, using two dial indicators (0.0001” resolution) 180 degrees apart from each other. Dowel rods (n=3) were mounted into our chuck and the spindle spun by hand. Runout was measured by subtracting the magnitude of one dial from the other. Runout was found to be under 3 thou for any given dowel, which matches what the manufacturer provides.

Actuation torque required is the amount of torque needed to spin Z and X handwheels under no load. This was measured by pushing the handle with a force gauge, perpendicular to the axis. Z axis was measured at 10N (1” radius handle) and X axis at 8N.

Summary of Lessons Learned

1. Define and draw out the structural loop for each subassembly before starting analysis.
2. Check that provided parts meet the specifications you care about. When they don’t, confirm the existence of reasonable off-the-shelf parts early in the design process.
3. Some functional requirements are directly from strict performance metrics, while others are set to aid in designing to meet those requirements. It is important to remember which of these can be revised throughout the process as you learn more about the machine you are building.
4. The HTM model for this lathe took approximately 100 man-hours. Perhaps 10% of that was learning how to use MathCAD, but the rest reflects the substantial time it can take to make an accurate model. We needed to check design changes against this model in the weeks that followed (which were answered within minutes). I don’t think these design changes were the real time-saving capabilities of the model. If we didn’t have a model, those decisions would be made from intuition or a “component-view” vs. “system-view” equations. The true time saving comes from building something that doesn’t work or not being able to confidently tell leadership why design choices are sound.
5. “Test what you want to test.” Even though we had many successful cuts on short (1-2”) pieces of aluminum and steel, the competition was for a relatively longer piece of aluminum, which our chuck failed to hold.
6. Understand how design decisions will potentially impact the assembly process, and how necessary changes to the assembly process may substantially impact other design decisions.

Appendix: Validation Procedures and Results

The following functional requirements were either not met or not measured. However, close investigation of the justifications behind these FR's and a holistic view on key performance metrics gives us confidence that the lathe we designed and built satisfies the course objectives and the needs of our model customer.

1. The requirement for total runout on a finished part was set to 4 micron +/- 4 however we measured 12 micron run out. Runout FR was set too aggressively under the false assumption that runout has an additive effect to error, counting against the repeatability specification. Since the lathe meets our repeatability FR, this runout FR can be relaxed.
2. The requirements for X, Y, and Z total machine stiffness were set in order to assist in meeting the repeatability requirement. Due to successful repeatability measurements, stiffness was not measured.
3. The requirement for minimum and maximum spindle RPM was set to aid in the cutting of steel and aluminum. The lowest measured spindle speed was 640 RPM, outside of the upper range (600 RPM) of this requirement. However, it is clear that the required speed range was set too wide in each direction, because good quality and high rate cuts can be accomplished within our achieved speed range. The highest measured spindle speed was 1400 RPM (no max speed requirement).
4. MTTF (mean time to failure) is typically done by measuring MTTF of components analytically, with FEA, or with a fatigue testing machine. We performed the first and second wherever possible; the last method was not possible given the scope of the class. Additionally, no wear or component failures were observed after extended use of the machine and all components were designed with life far exceeding this FR (18 months or 740 cutting hours).
5. Range of allowable operating temperatures was not measured due to practicality, but the measured performance on repeatability under laboratory temperatures allows for more than the budgeted error for the thermal error expected from the operating temperatures set in our Functional Requirement (12 - 28 °C).
6. Drop impulse load was not measured due to not wanting to damage our lathe but considerable design effort give us confidence that the lathe's primary components would be undamaged by a fall from table height.
7. Spindle impulse load, dynamic load capacity, and static load capacity were not measured numerically. However, the lathe performed well after tool crashes into the spindle, aggressive machining loads, and usage of the X and Z drives loaded under a person's full weight. These experiments satisfy the intent of this functional requirement.
8. The CNC controller (Universal G-code Sender) is consistently traveling 18-30 um short for any Z command (be it 3 mm or 20 mm). The error is not scaling linearly with distance traveled, therefore it is not a calibration issue. This may have to do with overcoming the static friction of our axis or be an issue with how commands are being sent.
9. 330 W (22A, 15.4V) were measured for a heavy steel cut. The motor does not stall out for a steel death-test of cutting from the center of a 5/8 " diameter stock. Likewise, for cutting power: the cutting power of the motor was estimated to be 337 W (assuming 90% efficiency). We did not stall out the motor for our heaviest steel cuts.
10. Actuation torque required is the amount of torque needed to spin the handle fo the Z and X axis. While our measured 0.3 Nm is technically under our specified range, the amount of torque to turn

these axes is satisfactory to the touch. We would not want to increase the torque needed, for fear of making the axes unresponsive.

11. Required actuation torque is “the maximum torque into X and Z handwheels which does not damage any components.” This is to ensure the machine is not so fragile that a heavy-handed user would break something. While this metric was not tested to failure, we were able to apply the required 6 Nm of torque without damage to any components.

Repeatability

Steel hex stock (½” diameter, n=5) were turned down using a single 50 thou radial depth of cut. A spring pass was used to reduce error from elastic deflection. A clutched digital micrometer was used to measure each diameter at four different locations. Repeatability was calculated as the difference between the maximum and minimum of the averages for each cylinder; concluding an overall repeatability of 31 microns.

Maximum: 9.744 mm
Minimum: 9.71325 mm
Difference: 0.031 mm

CNC Z Accuracy

The Z axis was commanded to move in 3.1mm increments along its range of motion. After each motion, a dial indicator (0.0005” resolution) was read to measure how much the axis traveled. The axis was also commanded to move 20 mm on a separate test, the error of that movement was not linearly higher compared to the smaller movements 3mm movements (both were on the order of 18-30 um). This is not a calibration issue.

commanded x movement -3.1mm (thou)	dial indicator reading (thou)	measured x movement (thou)	error on every movement (um)
0	-6.5	nan	nan
78.74	71.5	78	-18.796
157.48	149.5	78	-18.796
236.22	228	78.5	-6.096
314.96	305.5	77.5	-31.496
393.7	383.5	78	-18.796
472.44	462	78.5	-6.096
551.18	540	78	-18.796
629.92	618	78	-18.796
708.66	695.5	77.5	-31.496
		avg error	-18.796
		avg error /mm travel	-6.06

Fig. 1. CNC Z accuracy data

CNC X and Z Resolution

While CNC step resolution should theoretically be under 2 micron based on microstepping motor resolution and lead screw pitch, the minimum allowable commanded motion by our CNC controller (Universal G-code Sender) is 20 micron, enough to pass our functional requirement. The axes will not move for any step commanded below 20 microns.

CMM Tuning of X-Z Axes:

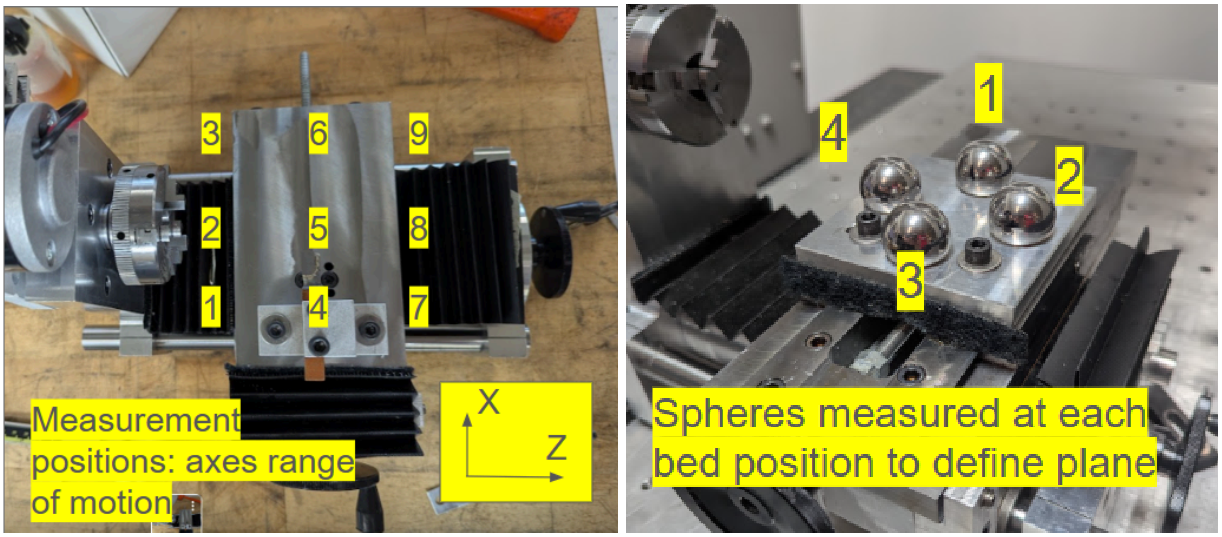

Fig. 2. CMM measurement positions

A plate with 4 spheres was attached to the female dovetail via the toolpost holder screws. The bed was moved to 9 locations across its range of motion. At each location, coordinate measurements of each sphere were taken. This defines a plane at each of the 9 positions. A mean plane and normal is calculated, and then each position's normal error can be calculated in each coordinate axis.

Maximum angle error: 0.0242 degrees
Average: 0.0155 degrees
StDev: 0.0065 degrees

Measurement Position (top view) (X up, Z right)

3	6	9
2	5	8
1	4	7

Total Angle Deviation (°)

0.0229	0.0242	0.0131
0.0123	0.0054	0.0077
0.0164	0.0178	0.0205

Z Angle Deviation (°)

-0.0146	0.0189	-0.0121
-0.0069	-0.0016	0.0009
0.0162	-0.0171	0.0163

X Angle Deviation (°)

0.0177	-0.0151	0.005
0.0101	-0.0051	-0.0077
0.0027	0.0049	-0.0126

Fig. 3. CMM normal vector deviations from mean normal. Colors are a relative scale from min to max

Then, each position's y position was calculated from the normal plane. The z-rails "slope downwards" as they get farther from the headstock, but this has minimal effect on lathe functional requirements. (Same measurement position layout as normal deviation, colors are a relative scale from minimum to maximum.)

Z Error from Mean Plane (thou)		
1.435	0.543	-1.193
0.673	0.639	-1.186
0.62	0.305	-1.837

Fig. 4. Error for each bed position in Z from the mean plane

Finally, the centroids of the measured points calculated at each of the 9 positions were used to calculate non-perpendicularity between the lathe x- and z-axes. This was used to adjust the dovetail-carriage bolted connection. Measurements after tuning are shown graphically below.

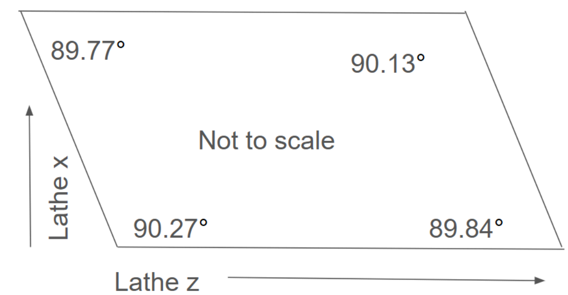

Fig. 5. Perpendicularity error between X and Z axes

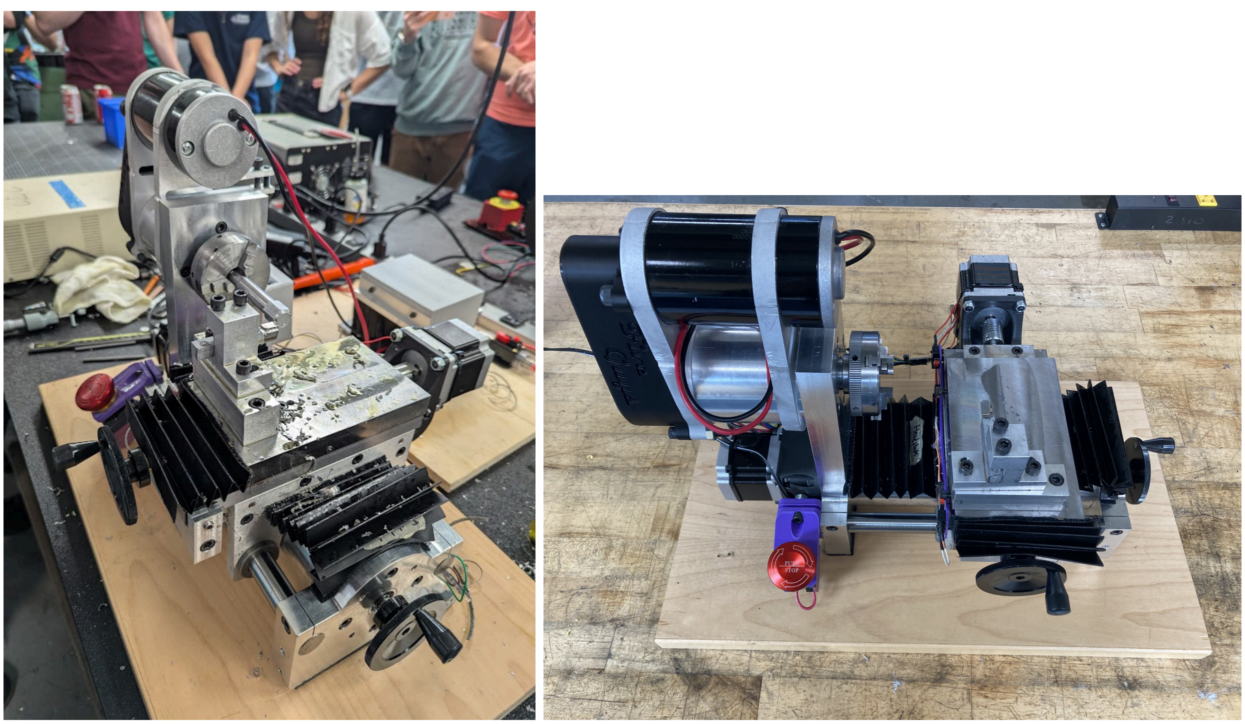

Fig. 6. Five Guys Lathe on Demo Day, 2025-05-12

Fig. 7. CNC machined aluminum part with 45 degree taper and concave radius